MRS Representation Issues

problems with unconventional MRSs

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Well-formed, Valid, and Conventional MRSs

- Consider an XML document:
  - It can be well-formed (follow XML syntax)
  - It can be valid (follow a schema)
- Now consider an MRS instance:
  - we have a definition (if a bit outdated) of well-formedness (e.g. for SimpleMRS)
  - but what about validity?
  - are there other conventions we should follow?

Conformance and Utility

- Ill-formed MRSs cannot be read/understood by MRS processors (like pyDelphin)
- Invalid MRSs are not guaranteed to work for operations like DMRS conversion
- Unconventional MRSs may surprise users and break assumptions, possibly leading to unintended consequences in MRS transformations

How do we make our semantic output valid/conformant/harmonious to maximize their convenience and utility for others?

Assumption: ARG0s are unique (except quantifiers)

```
[ LTOP: h0
  INDEX: e2 [ e TENSE: pres MOOD: indicative PROG: - PERF: - ASPECT:
  RELS: < [ "_akai_a_1_rel"<0:2> LBL: h4 ARG0: e2 ARG1: i3 ]
          [ "_naru_v_rel"<3:5> LBL: h1 ARG0: e2 ARG1: h4 ] >
  HCONS: < h0 qeq h1 > ]
```

aka-ku	naru
red-KU	become
"(it) redden(s)"	

see Jacy bug report #1

Assumption: ARG0s are "e" or "x" variables

```
[ LTOP: h0
  INDEX: e2 [ e TENSE: pres MOOD: indicative PROG: - PERF: - ASPECT:
  RELS: < [ "_rikai_s_1_rel"<0:2> LBL: h4 ARG0: e2 ARG1: i3 ARG2: i5
          [ rareru_v_can_rel<3:5> LBL: h4 ARG0: e2 ARG1: u6 ARG2: h7
          [ "_neg_v_rel"<6:8> LBL: h1 ARG0: i8 ARG1: h9 ] >
  HCONS: < h0 qeq h1 h7 qeq h4 h9 qeq h4 > ]
```

rikai	deki-nai
understanding	can-NEG

"(I) cannot understand." / "I don't get it"

see Jacy bug report #4

Assumption: ARG0s are "e" or "x" variables

Also in the ERG. E.g. "Kim unheard it."

```
[ LTOP: h0
  INDEX: e2 [ e SF: prop TENSE: past MOOD: indicative PROG: - PERF:
  RELS: < [ proper_q_rel<0:3> LBL: h4 ARG0: x3 [ x PERS: 3 NUM: sg I
          [ named_rel<0:3> LBL: h7 CARG: "Kim" ARG0: x3 ]
          [ "_hear_v_1_rel"<4:11> LBL: h1 ARG0: e2 ARG1: x3 ARG2: x9
          [ "_un-_a_rvrs_rel"<4:11> LBL: h1 ARG0: i10 ARG1: e2 ]
          [ pron_rel<12:15> LBL: h11 ARG0: x9 ]
          [ pronoun_q_rel<12:15> LBL: h12 ARG0: x9 RSTR: h13 BODY: h
  HCONS: < h0 qeq h1 h5 qeq h7 h13 qeq h11 > ]
```

Assumption: All EPs have an ARG0

```
[ LTOP: h0
  INDEX: e2 [ e TENSE: past MOOD: indicative PROG: - PERF: - ASPECT:
  RELS: < [ "discourse_x_rel"<-1:-1> LBL: h1 L-HNDL: h4 R-HNDL: h5 ]
          [ "_sorekara_x_rel"<0:4> LBL: h4 ARG0: e6 [ e TENSE: tense
          [ "_suru_v_rel"<5:6> LBL: h5 ARG0: e2 ARG1: i3 ARG2: i7 ] >
  HCONS: < h0 qeq h1 > ]
```

sorekara	shi-ta
after.that	do-PFV

"(I) did (it) after (that)."

see Jacy bug report #3

Surprising: Coindexed Underspecified Individuals

```
[ LTOP: h0
  INDEX: e2 [ e TENSE: tense MOOD: indicative PROG: - PERF: - ASPECT
  RELS: < [ "_sugiru_v_2_rel"<3:6> LBL: h1 ARG0: e4 [ e ...] ARG1: i
          [ "_taberu_v_1_rel"<0:2> LBL: h1 ARG0: e6 [ e ...] ARG1: i
          [ vv_rel<-1:-1> LBL: h1 ARG0: e2 ARG1: e6 ARG2: e4 ] >
  HCONS: < h0 qeq h1 > ]
```

tabe-sugiru

---

eat-exceed

"(I will) eat too much" / "overeat"

Other surprises and questions

- unattached (L)TOP
- variable properties on ICONS
- multiple constant args

Semantic Head-driven Spanning Tree

Given a notion of a semantic head (see next slide), can a single headed path (or tree) cover all EPs in a well-formed MRS?

What is a Headed Path?

Assume MRS is a connected graph (via arguments, hcons, and labels). Consider 2 EPs, A and B.

- A :RSTR/H> B, B is the head (A is a quantifier)
- A :ARG/EQ> B, B is the head (A is a modifier)
- A :ARG/NEQ> B, A is the head (B is A's argument)
- A :ARG/HEQ> B, A is the head (B is a scopal argument of A)
- A :ARG/H> B, A is the head (B is a QEQ argument of A)
- Difficult case: A and B share a label but aren't arguments of the other

"Nearly all dogs and cats bark loudly."

```
TOP:/H>"_bark_v_1_rel"(
  :ARG1/NEQ>_and_c_rel(
    :L-INDEX/NEQ>"_dog_n_1_rel"
      <RSTR/H:udef_q_rel &
    :R-INDEX/NEQ>"_cat_n_1_rel"
      <RSTR/H:udef_q_rel &
    <RSTR/H:_all_q_rel
      :/EQ:"_nearly_x_deg_rel")
  ) &
  <ARG1/EQ:"_loud_a_1_rel"
)
```

Missing slide: statistics of invalid structures and headed-path spanning trees in, e.g., Redwoods

Discussion Questions

1. Can we define a subset of *MRS that is "valid"?
2. Can we also define a subset of *MRS that is more conventional/harmonious with other grammars?
3. Where should the line lie between reasonable constraints and restrictions of expressive freedom?